SIDDAGANGA INSTITUTE OF TECHNOLOGY

Department of Electrical and Electronics Engineering
And
Department of Mechanical Engineering

MINIPROJECT PRESENTATION ON

"Fabrication and Testing of an Automated Underground Coal Mining Ventilation System"

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INTRODUCTION

- Coal mining is the process of extracting coal, a fossil fuel, from the earth using either surface mining (for shallower deposits) or underground mining (for deeper deposits).
- The underground mining is highly **hazardous** due to **toxic gases, dust, low oxygen, and risk of explosions**.
- Traditional ventilation systems use **manually operated fans and ducts**, which are **inefficient** and cannot **respond quickly** to changing conditions.
- Automatic ventilation system overcomes these problems by providing automated control over air quality and safety through sensors, data analysis, and fan adjustments.
- It uses **sensors, microcontrollers, and automatic operation and control of fan to provide** air quality and safety.**Gas and temperature sensors** continuously **monitor the environment**, while **fans are automatically regulated** to maintain **safe airflow**.
- This reduces **human intervention**, improves **safety**, saves **energy**, and ensures **reliable mining operations**.
- Thus, the **fabrication** of such a system enhances both **worker safety** and **operational efficiency**, and thus making mining **safer, smart and more sustainable**.

MOTIVATION

- Underground coal mines contain dangerous gases like methane and carbon monoxide that can rise quickly and become life-threatening.
- Traditional ventilation systems rely on manual monitoring, which is slow and cannot handle sudden gas concentration spikes.
- Delayed detection increases the risks of suffocation, explosions, and serious respiratory issues for miners.
- The reference highlights the need for real-time, automated gas monitoring to ensure continuous observation of air quality.
- An automated system can react instantly during emergencies, reduce human dependency, and significantly improve safety inside the mine.

LITERATURE SURVEY

Sl. NO.	Author(s)	Title	Source	Remarks
[1]	Gokuldas Nithesh Kumble, G. M. Akshay, M. K. Shreyas Gowda, K.	Fluid Flow Analysis of a Mine Ventilation Axial Fan Using CFD Techniques. In Conference on Fluid Mechanics and Fluid Power.	Singapore: Springer Nature Singapore , pp. 469–481, 2023.	• Provides valuable insights for optimizing mine ventilation fans using CFD analysis. • Ensures effective airflow distribution within underground mines. • Contributes to improved safety and better working conditions in mining systems.

LITERATURE SURVEY

Sl. NO.	Author(s)	Title	Source	Remarks
[2]	Huang, X., & Liu, Y.	Research and Design of Intelligent Mine Ventilation Construction Architecture	International Journal of Low-Carbon Technologies, Vol.17, pp.1232–1238 , 2022	• Provides a strong theoretical and technological foundation for mine ventilation. • Transforms traditional systems into intelligent and automated ones. • Enables adaptive control based on real-time mine conditions. • Utilizes data-driven approaches to enhance efficiency and safety.

LITERATURE SURVEY

Sl. NO.	Author(s)	Title	Source	Remarks
[3]	Xiaodong Lv	Research and Application of Intelligent Ventilation Control Systems in Coal Mines	Frontiers in Science and Engineering, Vol. 5, 2025.	• Focuses on upgrading coal mine ventilation systems. • Integrates intelligent sensing, control, and decision-making technologies. • Aims to enhance operational safety in mining environments. • Improves overall system efficiency through smart automation.

LITERATURE SURVEY

Sl. NO.	Author(s)	Title	Source	Remarks
[4]	Ihsan et al.,	Ventilation on Demand in Underground Mines Using Neuro-Fuzzy Systems	Engineering Applications of Artificial Intelligence (ACM Digital Library), Vol.133,Article no. 108048, 2024.	• Developed and tested a neuro-fuzzy control system for Ventilation on Demand (VoD). • Utilized sensor inputs and simulation/bench setups for validation. • Demonstrated that adaptive controllers reduce fan energy while maintaining safety thresholds. • Provided a practical architecture for integrating AI with PLCs and edge devices for real-world VoD deployment.

LITERATURE SURVEY

Sl. NO.	Author(s)	Title	Source	Remarks
[5]	S. S. Sutar, S. S. Patil, S. S. Kumbhar	Gas Monitoring & Ventilation System in Underground Coal Mine	International Journal of Creative Research Thoughts (IJCRT), Vol. 12 ,pp. 714, 2024.	• Presents a low-cost prototype for underground coal mine ventilation. • Uses Arduino Nano, MQ2 gas sensors, and flame detectors for system operation. • Demonstrates basic automation and sensor integration for safety monitoring. • Useful for early-stage fabrication, educational purposes, and proof-of-concept testing in mine safety systems.

LITERATURE SURVEY

Sl. NO.	Author(s)	Title	Source	Remarks
[6]	Hu Xiao, Wang Ning	Optimization and Intelligent Management of Coal Mine Ventilation Systems	Industrial and Mining Automation, , Vol.50, pp.33-34,56, (2024).	• Presents control strategies for coal mine ventilation using real-time monitoring and adaptive algorithms. • Utilizes centralized platforms for efficient system management. • Focuses on balancing safety and energy efficiency through intelligent regulation. • Provides practical guidance for designing and fabricating automated ventilation systems.

PROBLEM STATEMENT (GAP ANALYSIS)

- Underground coal mining plays a crucial role in meeting global energy demands, but it leads to the most hazardous working environment for miners.
- A major challenge is the lack of safe and adequate ventilation. In deep underground mines, harmful gases such as methane (CH_4), carbon monoxide (CO), and carbon dioxide (CO_2) are continuously released during coal extraction. If not properly diluted and removed, these gases can cause suffocation, poisoning, and even lead to catastrophic explosions.
- Traditional ventilation systems, which rely on manually controlled fans and duct arrangements, often fail to provide real-time adjustments to changing mine conditions. This manual approach is inefficient and slow to respond in emergencies, leads to life risk or reduces the life expectancy.
- Therefore, there is a demanding need for an **automated, reliable, and intelligent ventilation system** that can continuously monitor air quality, detect dangerous gas levels, and adjust quality airflow to ensure the safety and health of miners while maintaining energy efficiency.

OBJECTIVES

1. Phase – I Fabrication of coal underground mining & ventilation system:

Design and build a small-scale or prototype model of the underground coal mine ventilation system. It includes creating the mine layout, ducts, fans, sensors, and control units to simulate real time mining conditions for study and demonstration.

2. Phase – II Automation of ventilation system:

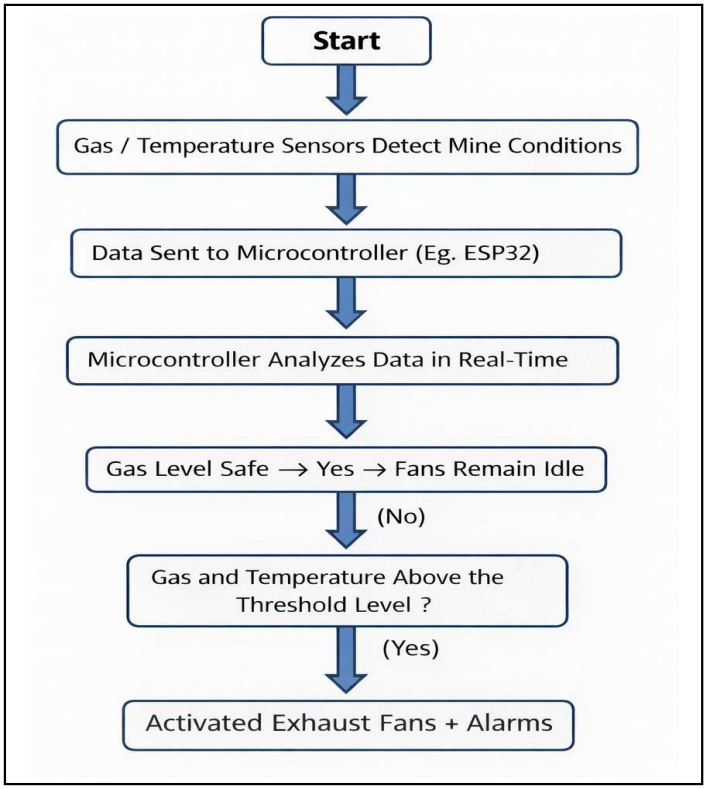
Automation includes sensors and controllers to make the ventilation system smart and self-operating. It can automatically detect gas concentration, temperature, or dust levels and adjust fan speed or airflow accordingly. This improves safety, efficiency, and reduces manual effort.

3. Phase – III Testing of fabricated model:

The developed model is tested to verify its working efficiency, accuracy, and safety. This system is tested for the working of ventilation system effectively circulates fresh air, removes harmful gases, and responds to control signals. This phase ensures reliability and sustainability before practical implementation.

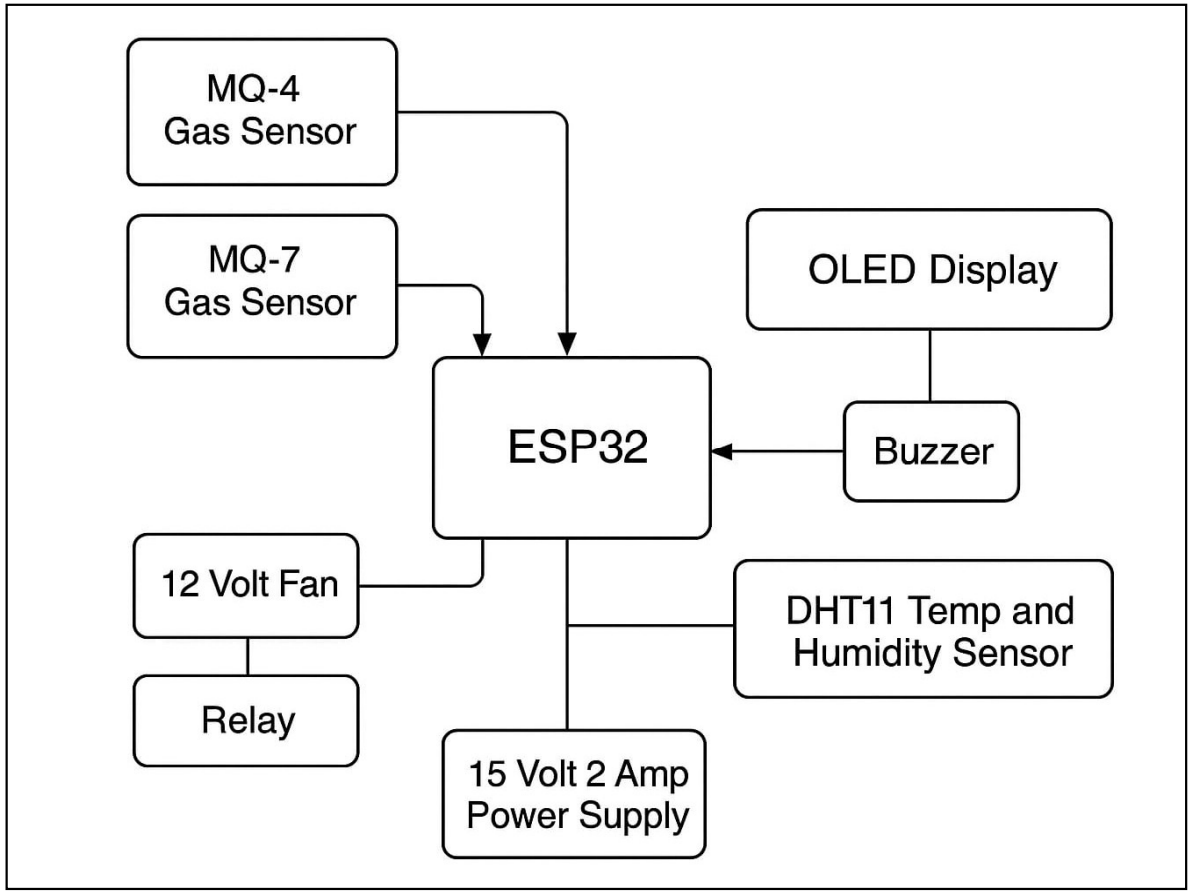
WORK FLOW

Ventilation is an essential component of underground mining, helps to maintain the safety and health of miners. Consequently, production and work efficiency can be enhanced. The principal purpose of the ventilation system is to regulate the quantity and quality of fresh air while eliminating hazardous gases



Fig(1):-Workflow of "Fabrication and Testing of an Automated Underground Coal Mining Ventilation System".

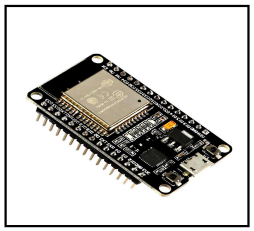
BLOCK DIAGRAM



Fig(2):-Block diagram of "Fabrication and Testing of an Automated Underground Coal Mining Ventilation System".

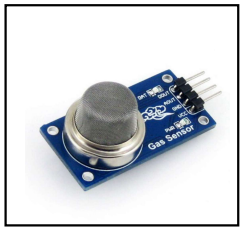
HARDWARE DESCRIPTION-I

- **ESP32 Microcontroller: ESP32 Microcontroller:** A dual-core 32-bit Wi-Fi and Bluetooth microcontroller with **3.3 V logic**, used as the main control unit for sensor data processing, display, and automatic fan and alarm control.



- **MQ-4 Gas Sensor:** Detects methane (CH_4) and natural gas in the range of 300-10000 ppm. Works on SnO_2 sensing element that changes resistance with gas concentration. Provides an analog signal to ESP32 for gas monitoring and leakage detection.

- **MQ-7 Gas Sensor:** Used to detect carbon monoxide (CO) gas from 20–2000 ppm. Operates at 5V using a SnO_2 sensor with variable resistance. Outputs analog voltage proportional to CO level for safety and air quality control.

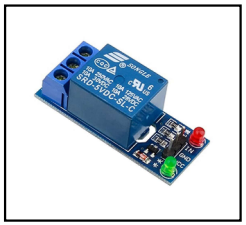


- **PVC**: PVC is used for ducts because it is lightweight, corrosion-resistant, and provides smooth airflow with easy installation and long-lasting durability. It is also cost-effective and suitable for both industrial and underground ventilation applications.

HARDWARE DESCRIPTION-II

- **12V DC Fan:** A small cooling fan operating at 12V and controlled by relay. Turns ON automatically through ESP32 when gas levels rise to ventilate the area and reduce concentration.

- **Relay Module:** An electromechanical 5V switch used to control high-power devices like fans or alarms. ESP32 sends a low-voltage signal to trigger the relay for automatic switching.

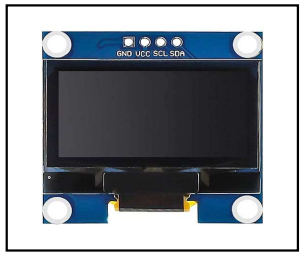


- **15V 2A Power Supply:**Provides 15V DC output from AC mains and powers all modules. Voltage regulators derive 5V for sensors and 3.3V for ESP32 ensuring stable operation of the system.

- **Acrylic Sheet**:An acrylic sheet is a lightweight, transparent plastic material known for its high strength, weather resistance, and excellent optical clarity. It is easy to cut, bend, and shape, making it ideal for displays, protective covers, and project enclosures.

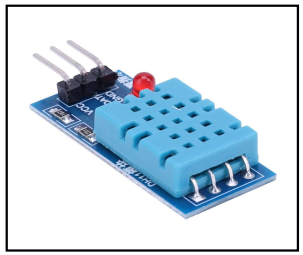
HARDWARE DESCRIPTION-III

- **OLED Display:** A 0.96" or 1.3" display (128×64 pixels) using I2C interface with low power consumption. Displays real-time data such as gas levels, temperature, and humidity received from ESP32.



- **Buzzer:** A 3.3V piezoelectric device used to give audible alerts. Activated by ESP32 when gas concentration or temperature exceeds safe limits to warn users immediately.

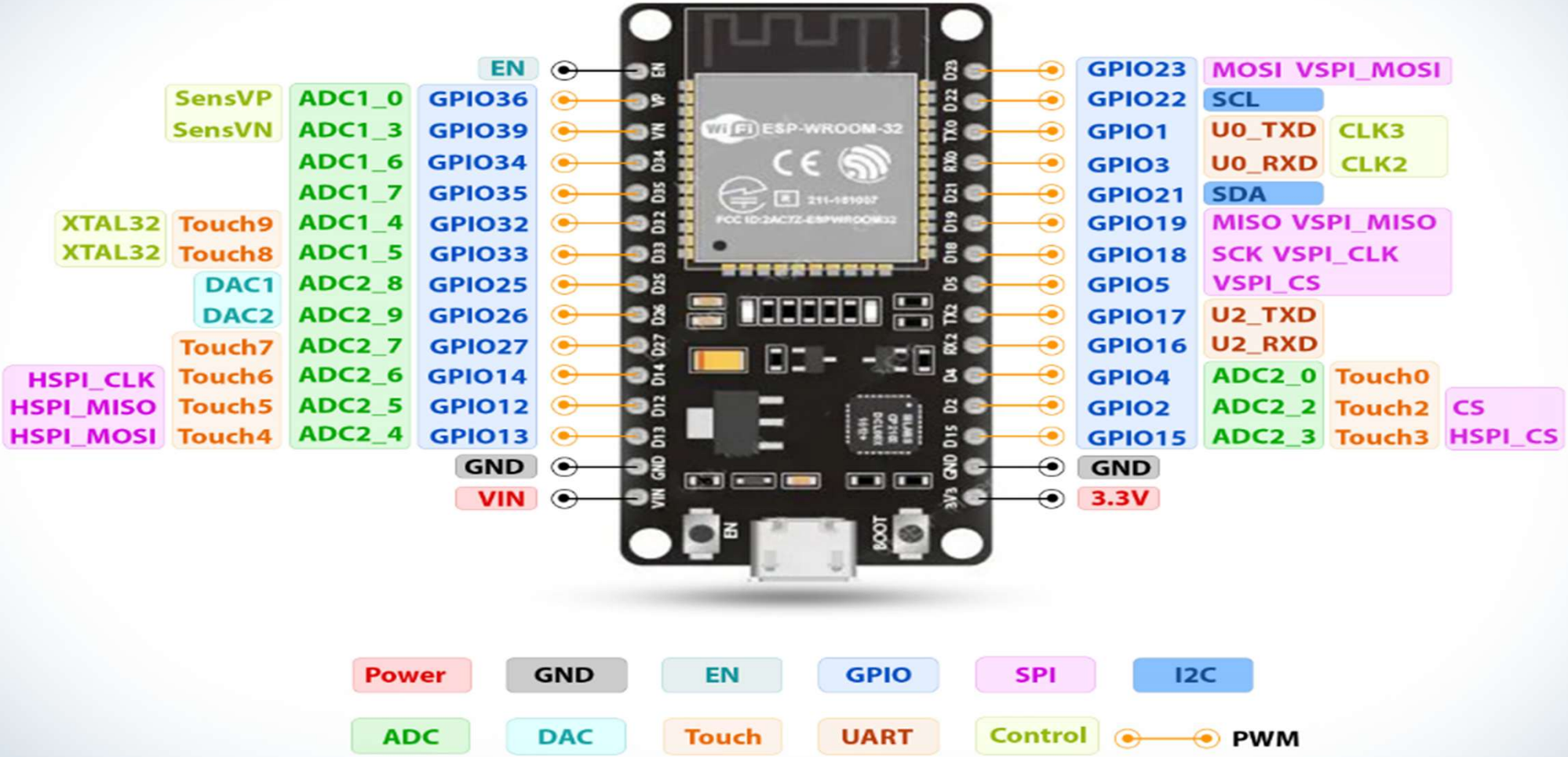
- **DHT11 Temperature & Humidity Sensor:** Digital sensor measuring 0–50°C temperature and 20–90% humidity. Sends data via single-wire communication to ESP32 for monitoring environmental conditions.



SOFTWARE DESCRIPTION

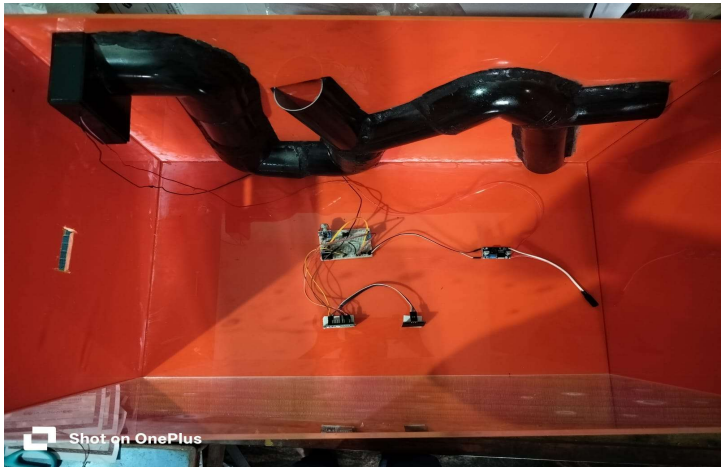
- The software is developed using **Arduino IDE** and uploaded to the **ESP32 microcontroller**.
- Reads real-time data from **MQ-4 (Methane)**, **MQ-7 (CO)**, and **DHT11 (Temperature & Humidity)** sensors.
- Processes sensor values and compares them with **predefined safety thresholds**.
- Automatically **activates the ventilation fan** through a relay when gas levels rise above safe limits.
- **Triggers the buzzer** to provide an immediate warning during hazardous conditions.
- Continuously updates the **OLED display** with methane level, CO level, temperature, humidity, and system status (SAFE/ALERT).
- Runs in a continuous loop to ensure **24/7 monitoring and fast response**.
- Includes basic **error detection** to identify faulty or disconnected sensors.
- Supports **serial monitoring** for testing, calibration, and debugging.
- Ensures reliable automation for improving **mine ventilation and worker safety**.

ESP32 DEV. BOARD PINOUT

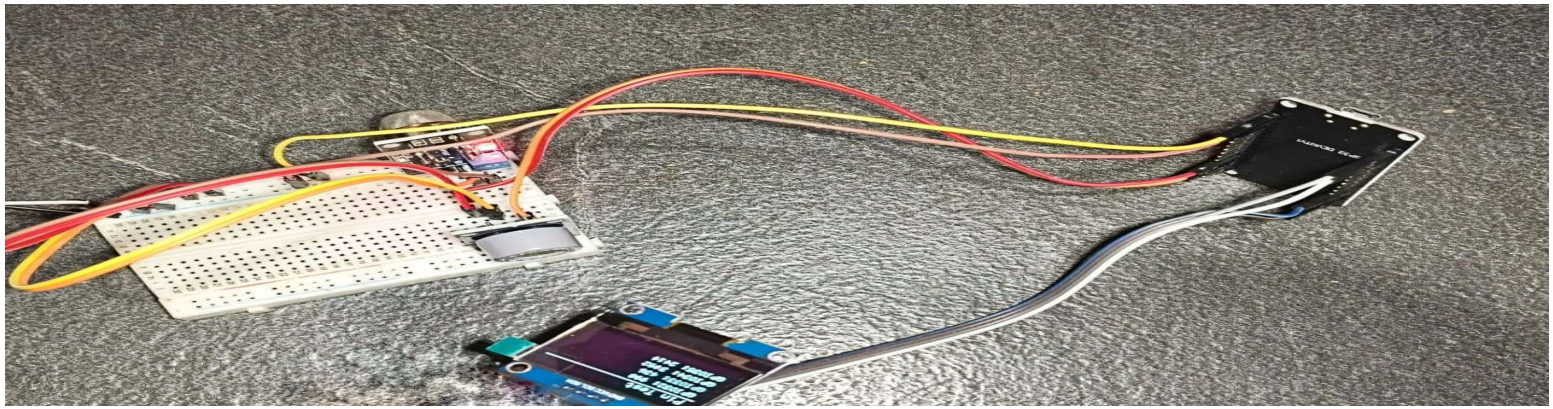


SNAPSHOTS

Fig(3):- Front view



Fig(4):- Top View



Fig(5):- Software View

BUDGET DETAILS

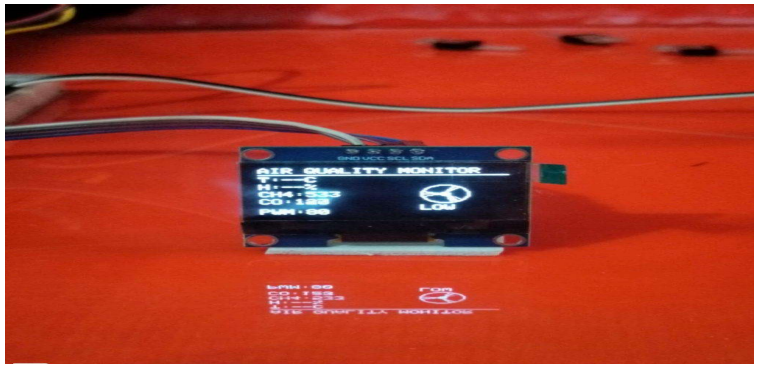
Components	Prices
Esp 32 microcontroller	₹600
Mq 4 gas sensor	₹187
Mq 7 gas sensor	₹259
Oled display	₹350
Buzzer	₹39
Dth 11 tmp and humidity sensor	₹160
12 volts fan	₹270
Relay	₹135
15volts 2 amps power supply	₹499
Miscellaneous	₹1000
Total:	₹3434

RESULTS

- The project successfully developed and demonstrated an automated coal mine ventilation system for continuous underground air-quality monitoring.
- MQ-4, MQ-7, and DHT11 sensors were integrated with an ESP32 microcontroller, a 12V ventilation fan, and alert mechanisms.
- The system detects methane and carbon monoxide in real time and monitors temperature and humidity.
- When gas levels exceed safe limits, the fan and buzzer activate automatically, and real-time readings are shown on the OLED display.
- Testing proved fast response, improved airflow, reduced manual intervention, and enhanced safety.
- The prototype demonstrates reliability, efficiency, and suitability for improving underground mining safety and sustainability.

RESULTS

PARAMETERS	SENSOR	SENSING RANGE	Threshold	OUTCOMES
METHANE /LPG	MQ-4	300-10000PPM	530ppm	600(buzz on)
CARBON MONOXIDE	MQ-7	20-2000PPM	300ppm	350(buzz on)
HUMIDITY	HUMIDITY SENSOR	0% to 100%	25	40(buzz on)
TEMPERATURE	Thermal sensor	0 to 100 °C	Above Room Temperature	Above Room Temperature



Fig(6):- Developed prototype of “Fabrication and Testing of an Automated Underground Coal Mining Ventilation System”.

CONCLUSION

- Automation ensures rapid response to dangerous gas levels, improving safety compared to manual ventilation methods.
- The integration of sensors and a microcontroller provides accurate, continuous air-quality assessment inside the mine.
- The system reduces the need for human intervention, lowering the risk to miners during hazardous situations.
- Overall, the automated ventilation setup offers a practical, reliable, and cost-effective solution for enhancing mine safety.
- The fabricated system effectively monitors methane, carbon monoxide, temperature, and humidity in real time.

SCOPE FOR FUTURE ENHANCEMENT

- Additional sensors for dust, toxic fumes, and airflow can improve detection accuracy and overall safety.
- Machine learning can be added to predict hazardous conditions and optimize ventilation using past data.
- IoT features like cloud storage, remote monitoring, and mobile alerts can enhance supervision and control.
- Automated calibration, predictive maintenance, and emergency protocols such as evacuation alerts or backup ventilation can increase reliability.
- A modular, scalable, and energy-optimized design can support larger mines and reduce power usage, making the system a fully intelligent and dependable ventilation solution.

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TASK	Week1 (Sep 8-14)	Week2 (Sep 24-30)	Week3 (Oct 5-11)	Week4 (Oct20-27)	Week5 (Nov5-11)	Week6 (Nov12 - 18)	Week7 (Dec 6-12)	Week8 (Dec13-19)
Finding a problem to address								
Discussion with guide for finalising								
Literature survey								
Developing basic design								
Selection of components								
Fabrication								
Testing and modification								
Report writing								
Report correction and finalising								

THANK YOU